

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T10 · Surds

Questions

Student book · questions with writing space

■ 14 pages

FAMILY MAP

Surds

F01	Rationalise a surd denominator	9 Q 29 marks
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F02	Simplify and collect surds	2 Q 4 marks
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F03	Multiply and expand surd expressions	2 Q 6 marks
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Working Unit 11

4MA1-1H Q15

UNIT 1
4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$$k = \dots\dots\dots$$

(1)

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(3)

4MA1-1H Q17

3 marks

Show each stage of your working clearly.

Question 27

4MA1-2H Q15 (b)

MODULAR UNIT 1

3 marks

(b) Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

.....
(3)

Question 27

4MA1-2H Q23

MODULAR UNIT 1

4 marks

23 The diagram shows a cuboid with a square cross section.

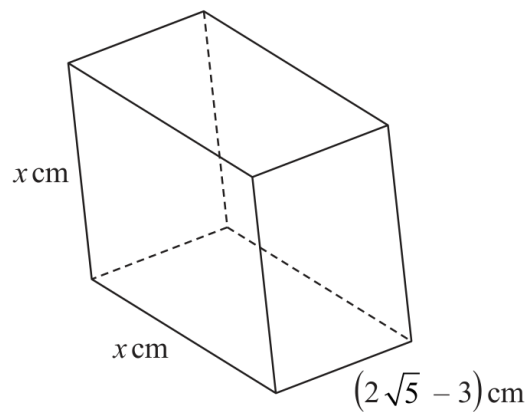


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

Question 13

4MA1-1H Q17 M01

MODULAR UNIT 1

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

Question 14

4MA1-1H Q17 M02

MODULAR UNIT 1

3 marks

- (b) Show that $\frac{5 - \sqrt{2}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(3)

4MA1-1H Q16 Q16

3 marks

Show your working clearly.

4MA1-2H Q17

3 marks

can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working.

The length of the rectangle is $(\sqrt{7} + 1)$ cm.

find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

Show each stage of your working clearly.

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)